

BISOCA, Romania

WMO: 152850

Lat: 45.5489N

Lon: 26.7103E

Elev: 851

StdP: 91.51

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.5	-11.2	-18.5	0.8	-9.4	-16.3	1.0	-7.9	23.6	-5.7	19.9	-6.8	5.8	0	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.9	26.2	18.5	24.9	17.9	23.7	17.3	20.1	24.7	19.0	23.5	18.1	22.4	3.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	14.6	23.0	17.2	13.6	22.0	16.3	12.8	20.9	61.5	24.7	57.7	23.6	54.7	22.6	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	13.8	11.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-2.4	-0.2	2.5	7.2	12.8	16.8	18.1	18.7	13.9	8.6	3.6	0.1		
	DBStd	8.61	5.42	5.44	5.20	5.21	3.47	3.21	3.08	3.36	4.42	4.96	5.22	5.13		
	HDD10.0	1649	383	288	238	113	14	1	1	0	15	88	201	308		
	HDD18.3	3757	642	520	491	333	173	67	43	39	142	303	441	565		
	CDD10.0	1047	0	1	5	30	101	203	250	268	133	44	10	1		
	CDD18.3	114	0	0	0	1	1	19	34	49	10	0	0	0		
	CDH23.3	339	0	0	0	1	7	44	109	153	26	0	0	0		
	CDH26.7	24	0	0	0	0	0	1	11	11	1	0	0	0		
(14) Wind	WSAvg	4.4	4.8	5.0	5.0	4.9	3.8	4.2	3.7	3.7	4.1	4.4	3.9	4.8		
(15) Precipitation	PrecAvg	700	31	27	35	55	92	103	91	85	53	49	38	38		
	PrecMax	937	72	78	67	109	185	181	201	168	105	168	76	88		
	PrecMin	447	8	3	3	27	30	35	29	22	12	4	0	5		
	PrecStd	139	18	17	17	25	44	38	45	39	28	36	21	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	14.9	17.4	21.1	23.7	26.1	28.0	28.0	26.3	20.5	17.1	14.1		
		MCWB	6.3	7.8	8.8	12.3	14.3	19.4	19.9	20.0	17.6	12.7	10.6	6.3		
	2%	DB	9.2	11.4	14.6	18.8	20.6	24.6	25.9	26.2	23.5	18.6	14.9	11.2		
		MCWB	4.6	6.1	8.1	11.4	14.1	18.5	18.9	18.1	16.1	12.1	8.8	4.9		
	5%	DB	7.7	9.0	11.9	17.0	19.2	23.2	24.4	25.0	21.6	17.2	12.7	8.7		
		MCWB	4.0	4.8	6.3	10.5	13.6	17.8	18.2	17.7	14.6	11.3	8.4	4.5		
	10%	DB	5.4	7.1	9.8	15.1	17.8	21.8	22.9	23.7	19.8	15.8	11.1	6.7		
		MCWB	2.5	3.7	5.5	8.9	12.8	16.8	17.4	17.1	14.1	10.7	7.2	3.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	9.0	10.2	13.4	16.5	20.8	21.5	21.4	19.1	14.2	11.4	7.7		
		MCDB	10.6	14.8	15.0	19.5	20.5	24.9	25.6	26.0	24.1	18.6	15.2	10.7		
	2%	WB	5.4	6.6	8.6	12.2	15.1	19.3	19.9	20.0	16.7	13.0	10.3	6.2		
		MCDB	8.7	10.6	13.4	17.6	19.0	23.5	24.5	24.8	21.8	17.1	13.7	10.3		
	5%	WB	4.3	5.4	7.1	11.1	14.1	18.3	18.8	18.8	15.7	12.1	9.2	5.1		
		MCDB	7.0	8.9	11.5	16.1	18.2	22.7	23.2	23.3	20.5	16.3	12.2	8.5		
	10%	WB	2.8	4.0	5.9	10.0	13.2	17.3	17.9	17.9	14.6	11.2	8.1	3.9		
		MCDB	5.3	6.6	9.4	14.7	17.1	21.6	21.8	22.4	19.3	14.8	10.6	6.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.1	6.4	6.7	6.5	6.7	6.6	6.9	6.1	5.7	5.6	5.2		
		MCDBR	7.0	8.2	9.0	8.2	7.7	7.6	8.1	7.7	7.4	6.8	7.2	7.0		
	5% WB	MCWBR	4.5	4.8	4.8	5.0	4.7	5.2	4.8	4.4	4.5	4.4	4.1	4.1		
		MCWBR	6.3	7.3	8.5	7.4	7.2	7.4	7.1	7.7	6.9	6.3	6.0	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.291	0.316	0.349	0.401	0.381	0.382	0.396	0.407	0.371	0.340	0.316	0.292		
	taud		2.398	2.333	2.316	2.195	2.317	2.335	2.299	2.262	2.340	2.421	2.448	2.417		
	Ebn at Noon		819	859	877	856	887	885	866	841	843	823	781	776		
	Edh at Noon		72	94	110	136	124	123	126	126	107	85	69	65		
(44) All-Sky Solar Radiation	RadAvg		1.29	2.11	3.26	4.46	5.61	5.95	6.06	5.48	3.98	2.59	1.42	1.09		
	RadStd		0.16	0.25	0.34	0.46	0.46	0.45	0.45	0.38	0.36	0.29	0.13	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (64 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	-121	-162	+75	+27

Nomenclature: See separate page